JEMAY SALOMON

Ph.D. Candidate in Quantitative Genetics

INRAE, GQE–Le Moulon, Université Paris-Saclay, 91190 Gif-sur-Yvette, France

+33 7 45 51 94 65 | jemay.salomon@inrae.fr | jemaysalomon.github.io

Summary

I am a PhD student (3rd year) in quantitative genetics. I'm interested in using advanced mixed-modeling approaches to better estimate genetic effects and (co)variance components (e.g. automatic differentiation using R/C++), evaluating model robustness through stochastic simulations, and, in a more general way, applying quantitative genetics theory to dissect the genetic architecture of complex traits. I'm also interested in simulation-based optimization of breeding-schemes, which includes implementing and comparing genomic and/or phenomic vs phenotypic selection, finding the right balance between genetic diversity and genetic gain/cost.

Education

Ph.D. in Quantitative Genetics **Oct. 2023 – present**
Université Paris-Saclay, Gif-sur-Yvette, FR

M.Sc. in Quantitative Genetics **Sept. 2021 – Sept. 2023**
Institut Agro Montpellier, Montpellier, FR

B.Sc. in Agronomy **Sept. 2013 – Sept. 2018**
Université Quisqueya, Port-au-Prince, HT

Professional experience

Research Assistant **Oct. 2023 – present**
INRAE – GQE Le Moulon, Gif-sur-Yvette, FR
- Investigate the genetic basis and architecture of interspecific mixing ability in wheat-pea association.
- Develop a joint breeding methodology for sole and intercropping systems.
- Design and implement research protocol.

Research Intern (Master II) **Mars 2023 – Aug. 2023**
CHIBAS, Mirebalais, HT
- Optimized genomic prediction models for sorghum yield across multiple environments
- Designed and implemented research protocol
- Collected and managed phenotypic data from field trials, contributing to high-quality dataset creation

Research Intern (Master I) **May 2022 – Aug. 2022**
CIRAD – CNRS, Montpellier, FR
- Modeled phyllochron-plastochron relationship in sorghum in order to predict panicle initiation date
- Supported field trial operations through phenotypic data collection
- Synthesized findings into a scientific report, presenting analytical results and growth model interpretations

Publications

Journal papers
- **Salomon, J.**, Enjalbert, J. & Flutre, T. (2026). Joint modeling of social genetic effects in mono- and pluri-specific groups : case study in intercrops. *Genetics*, iyag177. doi.org/10.1093/genetics/iyag177

Manuscripts in preparation
- **Salomon, J.**, Levardois, K., Freitas, V., Moutier, N., Enjalbert, J. & Flutre, T. (2026). Variability of social genetic effects in a wheat–pea intercropping trial over three environments. *In prep.*
- Charles, J.R., **Salomon, J.** & Pressoir, G. (2026). Genomic prediction of sorghum across environments : effects of training design and stability indices. *In prep.*

Software

- Flutre, T., Enjalbert, J., Forst, E., Remerand, M., & **Salomon, J**. (2026). **plantmix** : Genetic Study of Plant Mixtures (v1.0.2) [Logiciel R]. CRAN

Conference presentations

— **Salomon, J**., Levardois, K., Freitas, V., Moutier, N., Enjalbert, J. & Flutre, T. (2026).. Dissecting the genetic architecture of direct and social genetic values : case study in wheat-pea intercrops. *Présentation orale*, Mix4Agroecology. Paris, France.

— **Salomon, J**., Enjalbert, J., & Flutre, T. (2025). Leveraging a broad gradient of plant–plant interactions to efficiently breed for cereal–legume mixtures. *Poster*, XIX EUCARPIA Biometrics in Plant Breeding. Edinburgh,UK. hal-05546246

— **Salomon, J**., Enjalbert, J., & Flutre, T. (2025). Leveraging genetic diversity to study plant–plant interactions and breed for crop mixtures. *Poster*, PSL-Qlife Quantitative Biology Winter School 2025. Paris, France.

— **Salomon, J**., Enjalbert, J., & Flutre, T. (2024). Leveraging a broad gradient of plant–plant interactions to efficiently breed for cereal–legume mixtures. *Présentation orale*, SFE2 Conference. Lyon, France.

— **Salomon, J**., & Pressoir, G. (2023). Optimization of prediction models for yield stability in sorghum : physiological determinisms, genomic prediction incorporating G×E, and phenomic approaches. *Poster*, ELLS Scientific Student Conference 2023. Stuttgart, Germany.

Additional Training

Statistical Workshop **Nov. 2025**
Saint-Rémy-lès-Chevreuse, FR
Best Practices and Recent Developments in Nonlinear Mixed-Effects and Joint Longitudinal–Survival Modeling :
- Nonlinear mixed-effects models for biological and longitudinal data.
- Joint modeling of longitudinal and survival data.
- High-dimensional variable selection using LASSO and Bayesian approaches.
- Reproducible scientific programming and automatic differentiation.
- Statistical modeling applications in plant breeding using R and Python

BIOMETRICS EUCARPIA 2025 – Satellite Workshop **Sept. 2025**
The Roslin Institute, Edinburgh, UK
Running Simulation of Selective Breeding Programs Using AlphaSimR & FieldSimR
- Simulation and optimization of selective breeding programs using the AlphaSimR R package.
- Modeling and management of spatial heterogeneity in field trials using the FieldSimR R package.

Mentoring

- **Rouguiyatou Diallo** (04/2025–08/2025, co-supervised with T. Flutre) — Master's thesis topic : Characterize the genetic basis of phenotypic plasticity in wheat grown in solecrop versus in intercrop with pea, in terms of growth and epidemic limitation).
- **Rémi Viéban** (06/2026) — Internship topic (Engineer 1st year) : Evaluation of phenomic and genomic prediction in cereal–legume intercropping systems.

Skills

Breeding for intercropping	Quantitative genetics	Plant breeding
Statistical genetics	Linear Mixed Models	Satistical modeling
Genomic / phenomic selection	C++ Programming (intermediate)	R Programming

Languages

- Haitian creole (native or bilingual)
- French (native or bilingual)
- English (professional)

Interests

— Music Production : Managing a private record label

— DJ & Live performance

Referees

- **Dr Jérôme Enjalbert**, Research director (DR), GQE-Le Moulon (Université Paris-Saclay, INRAE, CNRS, AgroParisTech). jerome.enjalbert@inrae.fr : 01 69 15 68 55
- **Dr Timothée Flutre**, Research scientist (CR), GQE-Le Moulon (Université Paris-Saclay, INRAE, CNRS, AgroParisTech). timothee.flutre@inrae.fr : 01 69 15 61 33
- **Pr Dominique This**, Professor, Institut Agro Montpellier, UMR AGAP. dominique.this@supagro.fr : 04 67 61 58 29